

SmartShop — C4 Container Diagram

Current state · main branch · github.com/CSEN-SCU/csen-174-s26-team-project-smartshop

■ Person ■ Container (verified) ■ Unverified ■ Database ■ External service ■ Infrastructure

PERSON

■ **Browser User**

Budget-conscious grocery shopper comparing prices and travel distance across nearby stores

SMARTSHOP SYSTEM — prototypes/

Divya's Next.js App

Next.js 14 · TypeScript
Chat UI for grocery price queries.
Crisis detection gate before Gemini call. Deployed on Vercel.
api/chat/route.ts src/lib/db.ts
✓ verified

Caroline's Express App

Node.js · Express · JS
Finds cheaper product alternatives via AI. XSS-fixed renderAlternatives() uses textContent.
prototypes/Caroline/app.js
✓ verified

Shreeya's Prototype

Stack not confirmed
Located at prototypes/shreeya/.
AI API + DB per assignment requirements.
prototypes/shreeya/
■ unverified

Terry's Prototype

Stack not confirmed
Located at prototypes/terry/. AI API + DB per assignment requirements.
prototypes/terry/
■ unverified

SQLite Database

better-sqlite3 (local file) · Products, stores, prices, chat history ·
Used by Divya's app only
prototypes/divya/src/lib/db.ts
✓ verified

EXTERNAL SERVICES

Google Gemini API

GEMINI_API_KEY
LLM for grocery chat responses.
Called after crisis detection gate passes.
route.ts → Gemini client
✓ verified

OpenAI API

OPENAI_API_KEY
LLM for product alternative suggestions in Caroline's app.
prototypes/Caroline/app.js
✓ verified

Vercel

Hosting · GEMINI_API_KEY env var
Hosts Divya's Next.js app.
Auto-deploys on push to main.
vercel.app
✓ verified

GitHub Actions

CI on PR + push to main
Runs only Divya's db unit tests.
No coverage for Caroline, Shreeya, or Terry.
.github/workflows/ci.yml
✓ verified

KEY RELATIONSHIPS

From → To	Description	Call site
User → Divya's App	Chat message over HTTPS	route.ts
User → Caroline's App	POST /findAlternatives over HTTPS	app.js
Divya's App → Gemini	Chat completion (GEMINI_API_KEY) — after crisis gate	route.ts
Divya's App → SQLite	Price lookups + chat history writes	db.ts
Caroline → OpenAI	Chat completion (OPENAI_API_KEY)	app.js → renderAlternatives()
GitHub Actions → Tests	npm test in prototypes/divya/ only	ci.yml
Vercel → Divya's App	Build + deploy on push to main	—